

Problem 53. Let $P(x)$ be a polynomial of degree $n \geq 2$ with rational coefficients such that $P(x)$ has n pairwise different real roots forming an arithmetic sequence. Prove that among the roots of $P(x)$ there are two that are also the roots of some polynomial of degree 2 with rational coefficients.

Solution. Let

$$P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0.$$

Then the polynomial P has n distinct real roots $r_1 < r_2 < \cdots < r_n$ forming an arithmetic progression. By Vieta's theorem, we have $r_1 + r_2 + \cdots + r_n = -a_{n-1}$. Since a_{n-1} is rational, we infer that $\mu = \frac{r_1 + \cdots + r_n}{n}$ is rational as well.

Consider now polynomial $Q(x) = P(x + \mu)$. Since μ is rational, Q also has rational coefficients. Moreover, Q has roots $r'_1 < r'_2 < \cdots < r'_n$ with $r'_i = r_i - \mu$, which also form an arithmetic progression. We now consider two cases: either n is even or n is odd.

Case 1. n is odd. Since the mean of $r'_1, r'_2, \dots, r'_n$ is equal to 0, we can write

$$(r'_1, r'_2, \dots, r'_n) = \left(-\frac{n-1}{2} \cdot d, -\frac{n-3}{2} \cdot d, \dots, -d, 0, d, \dots, \frac{n-3}{2} \cdot d, \frac{n-1}{2} \cdot d \right)$$

for some real $d > 0$. Denoting $k = \frac{n-1}{2}$, again by Vieta's theorem we infer that

$$a_{n-2} = \sum_{-k \leq i < j \leq k} (id) \cdot (jd) = d^2 \cdot \sum_{-k \leq i < j \leq k} ij.$$

Observe that for all $1 \leq a < b \leq k$, summands $(-a) \cdot b$, $(-b) \cdot a$, $(-b) \cdot (-a)$ and $a \cdot b$ in the sum $\sum_{-k \leq i < j \leq k} ij$ cancel out, and the only summands that do not cancel out are of the form $(-a) \cdot a$ for $1 \leq a \leq k$. Hence

$$\sum_{-k \leq i < j \leq k} ij = - \sum_{1 \leq i \leq k} i^2.$$

Since $\sum_{1 \leq i \leq k} i^2$ is a positive integer, we have

$$d^2 = \frac{-a_{n-2}}{\sum_{1 \leq i \leq k} i^2},$$

hence d^2 is rational.

Consider now the quadratic polynomial $S(x) = (x - d)(x + d) = x^2 - d^2$. Observe that S has rational coefficients and both its roots are also distinct roots of Q . Hence, the polynomial $T(x) = S(x - \mu)$ is quadratic, has rational coefficients with the leading coefficient equal to 1, and its roots are $-d + \mu = r_{\frac{n-1}{2}}$ and $d + \mu = r_{\frac{n+3}{2}}$.

Case 2. n is even. Again, this means that we can write

$$(r'_1, r'_2, \dots, r'_n) = (-(n-1) \cdot d, -(n-3) \cdot d, \dots, -3d, -d, d, 3d, \dots, (n-3) \cdot d, (n-1) \cdot d)$$

for some real $d > 0$. Denoting $k = \frac{n}{2}$, by Vieta's theorem we infer that

$$a_{n-2} = \sum_{-k < i < j \leq k} (2i-1)d \cdot (2j-1)d = d^2 \cdot \sum_{-k < i < j \leq k} (2i-1)(2j-1).$$

A similar cancelling scheme as in Case 1 yields that

$$\sum_{-k < i < j \leq k} (2i-1)(2j-1) = - \sum_{1 \leq i \leq k} (2i-1)^2.$$

Again, $\sum_{1 \leq i \leq k} (2i-1)^2$ is a positive integer, hence

$$d^2 = \frac{-a_{n-2}}{\sum_{1 \leq i \leq k} (2i-1)^2}$$

is a rational. We may again consider the quadratic polynomial $T(x) = S(x - \mu)$ where $S(x) = (x - d)(x + d) = x^2 - d^2$, and conclude as in Case 1. $\square$

Problem 54. Let P be a cubic polynomial given by $P(x) = ax^3 + bx^2 + cx + d$, where a, b, c, d are integers and $a \neq 0$. Suppose that $xP(x) = yP(y)$ for infinitely many pairs x, y of integers with $x \neq y$. Prove that the equation $P(x) = 0$ has an integer root.

Solution. The equation $xP(x) = yP(y)$ can be rewritten as

$$a(x^4 - y^4) + b(x^3 - y^3) + c(x^2 - y^2) + d(x - y) = 0.$$

Since $x \neq y$, we can take out the factor $x - y$ to obtain

$$a(x^3 + x^2y + xy^2 + y^3) + b(x^2 + xy + y^2) + c(x + y) + d = 0.$$

Let $x + y = k$, i.e. $y = k - x$. Substituting this in the above equation, we get

$$a(k^3 - 2k^2x + 2kx^2) + b(k^2 - kx + x^2) + ck + d = 0.$$

Rewriting this as an equation in x , we have

$$(2ak + b)x^2 - (2ak^2 + bk)x + (ak^3 + bk^2 + ck + d) = 0. \quad (1)$$

If $2ak + b \neq 0$, then this is a quadratic equation. The discriminant is

$$\Delta = k^2(2ak + b)^2 - 4(2ak + b)(ak^3 + bk^2 + ck + d) = -4a^2k^4 - 8abk^3 - 3b^2k^2 - 8ack^2 - 4bck - 8adk - 4bd.$$

Since the leading coefficient $-4a^2$ is negative, there are finitely many integers k such that $\Delta \geq 0$. For each such k , there are at most 2 solutions for x in (1). As it is given that there are infinitely many pairs of x, y (hence x, k) satisfying this equation, we must have $2ak + b = 0$ for some solutions. In that case, (1) becomes $ak^3 + bk^2 + ck + d = 0$. As it has a solution, the integer k is a root of $P(x) = 0$. $\square$

Problem 55. Let f be a function from the set of integers to the set of positive integers. Suppose that for any two integers m and n , the difference $f(m) - f(n)$ is divisible by $f(m - n)$. Prove that for all integers m, n with $f(m) \leq f(n)$ the number $f(n)$ is divisible by $f(m)$.

Solution. Suppose that x and y are two integers with $f(x) < f(y)$. We will show that $f(x) \mid f(y)$. By taking $m = x$ and $n = y$ we see that

$$f(x - y) \mid |f(x) - f(y)| = f(y) - f(x) > 0,$$

so $f(x - y) \leq f(y) - f(x) < f(y)$. Hence the number $d = f(x) - f(x - y)$ satisfies

$$-f(y) < -f(x - y) < d < f(x) < f(y).$$

Taking $m = x$ and $n = x - y$ we see that $f(y) \mid d$, so we deduce $d = 0$, or in other words $f(x) = f(x - y)$. Taking $m = x$ and $n = y$ we see that $f(x) = f(x - y) \mid f(x) - f(y)$, which implies $f(x) \mid f(y)$. $\square$

Problem 56. In each of six boxes $B_1, B_2, B_3, B_4, B_5, B_6$ there is initially one coin. There are two types of operations allowed:

Type 1: Choose a nonempty box B_j with $1 \leq j \leq 5$. Remove one coin from B_j and add two coins to B_{j+1} .

Type 2: Choose a nonempty box B_k with $1 \leq k \leq 4$. Remove one coin from B_k and exchange the contents of (possibly empty) boxes B_{k+1} and B_{k+2} .

Determine whether there is a finite sequence of such operations that results in boxes B_1, B_2, B_3, B_4, B_5 being empty and box B_6 containing exactly $1000^{1000^{1000}}$ coins. (Note that $a^{b^c} = a^{(b^c)}$.)

Solution. Yes. Denote by $(a_1, a_2, \dots, a_n) \rightarrow (a'_1, a'_2, \dots, a'_n)$ the following: if some consecutive boxes contain $a_1, \dots, a_n$ coins, then it is possible to perform several allowed moves such that the boxes contain $a'_1, \dots, a'_n$ coins respectively, whereas the contents of the other boxes remain unchanged.

Let $A = 1000^{1000^{1000}}$. Our goal is to show that

$$(1, 1, 1, 1, 1, 1) \rightarrow (0, 0, 0, 0, 0, A).$$

First we prove two auxiliary observations.

Lemma 1. $(a, 0, 0) \rightarrow (0, 2^a, 0)$ for every $a \geq 1$.

Proof. We prove by induction that $(a, 0, 0) \rightarrow (a - k, 2^k, 0)$ for every $1 \leq k \leq a$. For $k = 1$, apply Type 1 operation to the first box:

$$(a, 0, 0) \rightarrow (a - 1, 2, 0) = (a - 1, 2^1, 0).$$

Now assume that $k < a$ and the statement holds for some $k < a$. Starting from $(a - k, 2^k, 0)$, apply Type 1 operation to the middle box 2^k times, until it becomes empty. Then apply Type 2 operation to the first box:

$$(a - k, 2^k, 0) \rightarrow (a - k, 2^k - 1, 2) \rightarrow \dots \rightarrow (a - k, 0, 2^{k+1}) \rightarrow (a - k - 1, 2^{k+1}, 0).$$

Hence,

$$(a, 0, 0) \rightarrow (a - k, 2^k, 0) \rightarrow (a - k - 1, 2^{k+1}, 0).$$

$\square$

Lemma 2. For every positive integer n , let $P_n = \underbrace{2^{2^{\cdot^{\cdot^{\cdot^2}}}}}_n$ (e.g. $P_3 = 2^{2^2} = 16$). Then $(a, 0, 0, 0) \rightarrow (0, P_a, 0, 0)$ for every $a \geq 1$.

Proof. Similarly to Lemma 1, we prove that $(a, 0, 0, 0) \rightarrow (a - k, P_k, 0, 0)$ for every $1 \leq k \leq a$.

For $k = 1$, apply Type 1 operation to the first box:

$$(a, 0, 0, 0) \rightarrow (a - 1, 2, 0, 0) = (a - 1, P_1, 0, 0).$$

Now assume that the lemma holds for some $k < a$. Starting from $(a - k, P_k, 0, 0)$, apply Lemma 1, then apply Type 1 operation to the first box:

$$(a - k, P_k, 0, 0) \rightarrow (a - k, 0, 2^{P_k}, 0) = (a - k, 0, P_{k+1}, 0) \rightarrow (a - k - 1, P_{k+1}, 0, 0).$$

Therefore,

$$(a, 0, 0, 0) \rightarrow (a - k, P_k, 0, 0) \rightarrow (a - k - 1, P_{k+1}, 0, 0).$$

□

Now we prove the statement of the problem.

First apply Type 1 operation to box 5, then apply Type 2 operation to boxes B_4, B_3, B_2 and B_1 in this order. Then apply Lemma 2 twice:

$$\begin{aligned} (1, 1, 1, 1, 1, 1) &\rightarrow (1, 1, 1, 1, 0, 3) \rightarrow (1, 1, 1, 0, 3, 0) \rightarrow (1, 1, 0, 3, 0, 0) \rightarrow (1, 0, 3, 0, 0, 0) \rightarrow (0, 3, 0, 0, 0, 0) \\ &\rightarrow (0, 0, P_3, 0, 0, 0) = (0, 0, 16, 0, 0, 0) \rightarrow (0, 0, 0, P_{16}, 0, 0). \end{aligned}$$

We already have more than A coins in box B_4 , since

$$A = 1000^{1000^{1000}} < (2^{10})^{1000^{1000}} = 2^{10 \cdot 1000^{1000}} < 2^{1000^{1001}} < 2^{(2^{10})^{1001}} = 2^{2^{10 \cdot 1001}} < 2^{2^{2^{14}}} < P_{16}.$$

To decrease the number of coins in box B_4 , apply Type 2 operation to this box repeatedly until its size decreases to $A/4$. (In every step, we remove a coin from B_4 and exchange the empty boxes B_5 and B_6 .)

$$(0, 0, 0, P_{16}, 0, 0) \rightarrow (0, 0, 0, P_{16} - 1, 0, 0) \rightarrow (0, 0, 0, P_{16} - 2, 0, 0) \rightarrow \cdots \rightarrow (0, 0, 0, A/4, 0, 0).$$

Finally, apply Type 1 operation repeatedly to empty boxes B_4 and B_5 :

$$(0, 0, 0, A/4, 0, 0) \rightarrow \cdots \rightarrow (0, 0, 0, 0, A/2, 0) \rightarrow \cdots \rightarrow (0, 0, 0, 0, 0, A).$$

□